



Problems (Partial Derivatives)

- Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for the following functions
 - $f(x, y) = (x^2 - 1)(y + 2)$
 - $f(x, y) = e^{x+y+1}$
- If z is a function of x and y such that $xy - yz = xz = 0$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$
- If $z = \ln(\sqrt{x^2 + y^2})$, show that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1$
- Show that $f(x, y) = \ln(x^2 + y^2)$ satisfies $f_{xx} + f_{yy} = 0$
- If $u = x^2 - 2y^2 + z^3$ and $x = \sin t$, $y = e^t$, $z = 3t$, find $\frac{du}{dt}$
- The value of an equity forward is given as $V(S_0, r, T) = S_0 - Ke^{-rT}$. Develop the PnL equation of the forward position for small changes in the underlying risk factors.